Particle Physics Homework 3

1) EM gauge field

Consider $\mathcal{L}(A^\mu, \partial^\mu A^\nu) = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - j^\mu A_\mu$

where $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. We have learned in the lecture that the equation of motion is equivalent to 2 Maxwell equations.

1.1) Using $F^{\mu\nu}$ property to show the Bianchi identity

$$\partial_\mu F_{\nu\lambda} + \partial_\lambda F_{\mu\nu} + \partial_\nu F_{\lambda\mu} = 0$$

1.2) Show that the identity above is equivalent to another 2 Maxwell equations

1.3) Find the Hamiltonian of this field

2). Massive gauge field

Consider $\mathcal{L}(A^\mu, \partial^\mu A^\nu) = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + m^2 A_\mu A^\mu - j^\mu A_\mu$

where $F^{\mu\nu} \equiv \partial^\mu A^\nu - \partial^\nu A^\mu$

2.1) Find the equation of motion of this field

2.2) Find the Hamiltonian of this field

2.3) Show that Lorentz transformation is a symmetry of the massive gauge field, i.e.,

$$A^\mu \to A'^\mu = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x)$$

3) Energy-momentum tensor of gauge field

Consider $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$

3.1) Show that the energy momentum tensor is given by

$$T^{\mu}{}_{\nu} = -F^{\mu\rho}\partial_{\nu}A_{\rho} + \frac{1}{4}F^{\rho\sigma}F_{\rho\sigma}\delta^{\mu}{}_{\nu}$$

which is not symmetric in $\mu \leftrightarrow \nu$

3.2) Show that we can choose

$$T'^{\mu}{}_{\nu} = T^{\mu}{}_{\nu} + \partial_{\lambda}K^{\lambda\mu}{}_{\nu}$$

where $K^{\lambda\mu}{}_{\nu} = F^{\mu\lambda}A_{\nu}$

to make $T'^{\mu}{}_{\nu}$ symmetric

3.3) Show that

$$T'^{0}{}_{0} = \frac{1}{2}(\vec{E}^2 + \vec{B}^2) = \text{energy density}$$

$$T'^{0}{}_{i} = \varepsilon_{ijk}E_jB_k = (\vec{E}\times\vec{B})_i$$

$$= \text{Poynting vector}$$